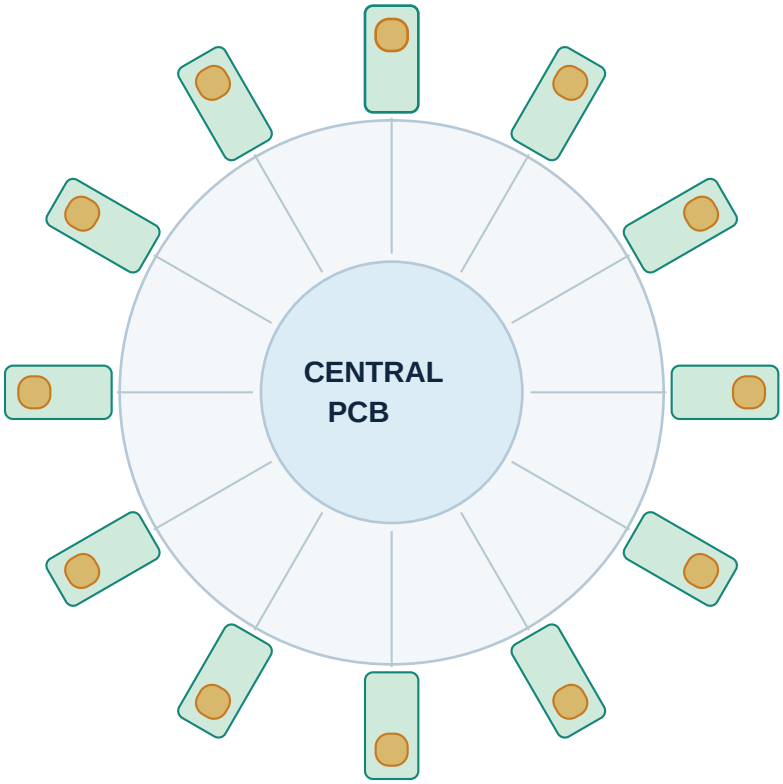
UCA-12 | Current Design Plan

12-element radial uniform circular array / 2.4 GHz / Revision 2 / 19 Oct 2026

PROJECT OVERVIEW

- 12 identical outward-facing antenna boards arranged at 30-degree azimuth intervals.
- One central phased-array-controller-v1 PCB for phase control, RF distribution, and MCU.
- Intended electronic azimuth selection across 360 degrees; approx. 30-degree sectors are a target, not a measured beamwidth.
- Bidirectional TX/RX is a design goal. RF switching and common TX/RX port routing remain open.



ARRAY GEOMETRY

12 elements / one disk

30 degrees between modules

Radially outward-looking

RF center near 2.4 GHz

Approx. free-space lambda: 125 mm

Half-wavelength: about 62.5 mm

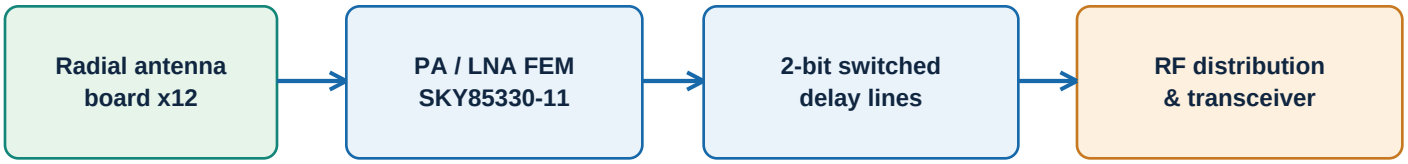
SELECTED COMPONENTS / CURRENT ASSUMPTIONS

RF switch	**SKY13453-385LF**	4 per element; 48 needed, 50 planned
RF front-end	**SKY85330-11**	1 per element; 12 installed, 13 ordered
MCU	**ESP32-S3 (planned)**	24 phase bits plus FEM control
Combining	**12:1 passive RF network**	Wilkinson tree concept; topology TBD

RF Architecture

Antenna-element-v1 + central phased-array-controller-v1

TOP-LEVEL SIGNAL PATH (CONCEPTUAL)



Per antenna PCB

Central PCB, 12 copies

RECEIVE PATH

Antenna -> FEM antenna port -> FEM LNA output -> phase-shift network -> passive combiner -> receiver.

Low-noise amplification occurs on each antenna board before cable, switch and combiner losses.

A passive 12-way combiner has about 10.8 dB ideal combining/splitting loss per independent input path; coherent combining changes the net directional result.

TRANSMIT PATH

Transmitter -> passive splitter -> phase-shift network -> FEM PA input -> FEM antenna port -> antenna.

Each FEM exposes distinct TX and RX RF connections; the center board needs a suitable routing arrangement to reuse the same phase-shift network.

TX output powers, combining/splitting loss, PA linearity, antenna gain, and regulatory EIRP must be evaluated before transmission.

REQUIRED ROUTING DECISION

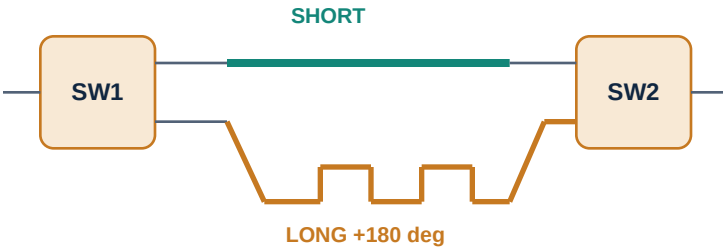
Do not connect the FEM TX and RX pins together directly. Investigate an appropriate RF SPDT switch per channel, an RF circulator/duplexing alternative, or independent TX/RX paths. Verify the FEM pinout and control truth table first.

Two-Bit Phase Shifter

One of 12 channels / 4 x SKY13453 SPDT switches per channel

ORIGINAL FOUR-SWITCH RF ROUTING

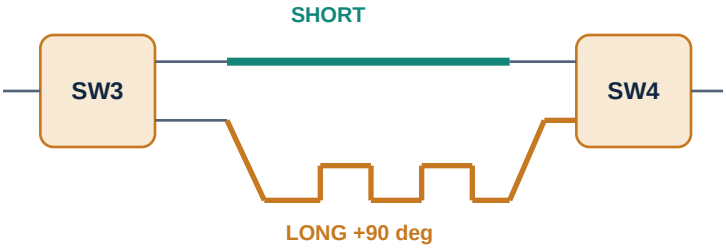
STAGE A | 180-DEGREE BIT



Paired switches share phase-bit selection

SPDT pair selects one of two lengths

STAGE B | 90-DEGREE BIT



Paired switches share phase-bit selection

Second SPDT pair adds another delay

PHASE STATES (IDEAL AT DESIGN FREQUENCY)

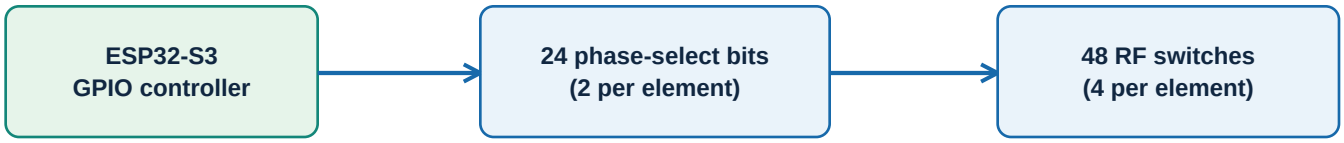
00 = 0 deg	**01 = 90 deg**	**10 = 180 deg**	**11 = 270 deg**

Actual RF phase is set by the difference in electrical length, including switch parasitics and PCB dielectric. A 90-degree delay is roughly one-quarter guided wavelength; the drawing is not a routed PCB or validated schematic.

Control, Budget & Next Steps

Open items and validation gates before PCB release

DIGITAL CONTROL



Each phase bit drives a matched pair of SPDTs, one at each end of a selected delay route. SKY13453 VCTL high is rated to 3.0 V maximum (and no higher than VDD); 3.3 V ESP32-S3 GPIO must use appropriate level translation. Dedicated FEM TX/RX enables also require GPIO.

UPDATED SWITCH PRICING / ESTIMATED PARTS

50 x SKY13453 @ $0.4856 (DigiKey 25+ tier)	**$24.28**
13 x SKY85330 @ $3.21 (user-stated tier)	**$41.73**
Illustrative parts subtotal (before fees)	**$66.01**
Previous tariff + shipping: NOT recalculated	**$18.20**

The old $103.20 cart included SKY13351 switches and is superseded; these new values are estimates, not a checkout quote. Taxes, tariff and shipping need a refreshed cart. PCB, matching, RF routing, combiner, connectors, and testing are excluded.

DESIGN GATES / OPEN QUESTIONS

- [ ] Obtain full SKY85330-11 electrical/pin details and define safe shared TX/RX routing.
- [ ] Validate SKY13453 2.8 V supply and translate ESP32 3.3 V logic to <= 3.0 V VCTL; simulate 50-ohm switched-delay networks.
- [ ] Check antenna pattern overlap for 12 outward-facing modules; 30-degree sectors are not guaranteed.
- [ ] Design/characterize the 12-way passive combiner and TX power splitter, including losses.
- [ ] Plan RF calibration, phase tables, test connectors and a VNA-based bring-up procedure.
- [ ] Validate TX power, EIRP, PA bias sequencing, thermal design, and local regulations.

Sources & Design Notes

Primary component documentation and scope notes

COMPONENT DATA

Skyworks SKY13453-385LF datasheet

https://www.skyworksinc.com/-/media/SkyWorks/Documents/Products/1901-2000/SKY13453-385LF_202830G.pdf

Skyworks SKY85330-11 product summary

https://www.skyworksinc.com/-/media/SkyWorks/Documents/Products/2801-2900/SKY85330_11_204868A_PS.pdf

DigiKey SKY13453-385LF listing

https://www.digikey.com/en/products/detail/skyworks-solutions-inc/SKY13453-385LF/5214219

DigiKey SKY85330-11 listing

https://www.digikey.com/en/products/detail/skyworks-solutions-inc/SKY85330-11/16579819

CONCEPT STATUS

- This is a captured design direction, not a fabrication-ready schematic or verified RF design.

- Selected switch changed to SKY13453-385LF; costs reflect a new estimate, not the previous $103.20 shopping cart. Inventory and quotes may change.

- The phase-shifter sketch is conceptual; four SPDT switches give two binary delay choices, not independently controlled 48 bits.

- The central board is currently named phased-array-controller-v1; antenna board is antenna-element-v1.

- SKY13453 needs VDD 2.5-3.5 V and VCTL <= 3.0 V and <= VDD: do not connect ESP32 3.3 V GPIO directly. Verify phase states and control truth table. Follow RF emission rules before TX.